

XGBoost Regressor

Dataset

House	Size	Age	price
H ₁	1000	20	50
H ₂	1200	15	55
H ₃	1500	10	70
H ₄	1800	5	85

Step 1: Initial prediction

$$\text{mean target} = (50 + 55 + 70 + 85) / 4 = 65$$

Step 2: Calculate gradient

House	y_i Actual	$\hat{y}_i$ prediction	$g_i = \hat{y} - y_i$ Gradient
H ₁	50	65	+15
H ₂	55	65	+10
H ₃	70	65	-5
H ₄	85	65	-20

Step 3: Hessians

For squared error

$h_i = 1$ for every row

Step 4: Find first split

try size < 1400

Left Node

H₁, H₂ = 15, 10

Sum (G_L) = 25

$H_L = (1+1) = 2$

Right Node

H₃, H₄ = -5, -20

$G_R = -25$

$H_R = 2$

Step 5: Calculate Gain = $\frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma$

$$\text{Gain} = \frac{1}{2} \left(\frac{(25)^2}{2+1} + \frac{(-25)^2}{2+1} - \frac{(0)^2}{5} \right) - \gamma$$

assuming $\lambda = 1$

$$\gamma = 0$$

$$208.3$$

Good split

Step 6 Split left node again

House

Gradient

H_1

15

H_2

10

Try split

Age < 18

left-left node

$$H_2(G_L) = 10$$

$$\text{Hessian} = 1$$

left-Right node

$$H_1(G_R) = 15$$

$$\text{Hessian} = 1$$

$$\text{Gain} = \frac{1}{2} \left(\frac{(10)^2}{1+1} + \frac{(15)^2}{1+1} - \frac{(10+15)^2}{1+1+1} \right) - \gamma$$

$$\frac{1}{2} (50 + 112.5 - 208.3) - 0$$

$$-22.91$$

Bad Split

Try split left Node
Age < 8

Right left

$$H_4 = G = -20$$

Right - Right

$$H_3 = G = -5$$

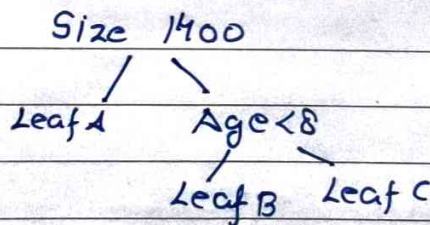
$$\text{Gain} = \frac{1}{2} \left(\frac{(-20)^2}{1+1} + \frac{(-5)^2}{1+1} - \frac{(-20-5)^2}{1+1+1} \right) - \gamma$$

$$\frac{1}{2} (200 + 12.5 - 208.3) - 0$$

$$2.08$$

+ve split

Final tree



Step 8 - Compute leaf value

$$\text{leaf A} = G = 25$$

$$H_2 = 2$$

$$W = \frac{-G}{H+1} = \frac{-25}{2+1} = -8.33$$

$$\text{leaf B} = \frac{-(-20)}{1+1} = 10$$

$$\text{leaf C} = \frac{-(-5)}{1+1} = 2.5$$

Update prediction

$$H_2, H_1 = F_0 + \eta F_1(00)$$

$$\eta = 0.1$$

$$65 + 0.1(-8.33)$$

$$64.17$$

$$H_3 = 65 + 0.1(2.5)$$

$$65.25$$

$$H_4 = 64 + 0.1(10)$$

$$= 66$$

House

Prediction

H_1

64.17

H_2

64.17

H_3

65.25

H_4

66

find gradient repeat the process